

# BYIRINGIRO Benit Gilbert

Software Engineering Graduate | Full-Stack & ML-Integrated Systems Developer

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## PROFESSIONAL SUMMARY

Driven Software Engineering graduate from UNILAK with a strong portfolio of complex, production-grade systems built independently. Developed Abelus — an enterprise multi-tenant commerce and POS platform — and LaptopLifeSaver — an AI-powered predictive maintenance system with a real-time fleet dashboard and Python monitoring agent. Demonstrates solid command of full-stack development, machine learning integration, and system architecture. Ready to contribute immediately in a professional internship environment.

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## EDUCATION

**Bachelor of Science in Software Engineering (BSE)** | University of Lay Adventist of Kigali (UNILAK) 2021 – 2025

- Coursework: Software Development, Data Structures & Algorithms, Database Systems, Web Engineering, OOP
- Built multiple production-grade software systems as part of academic and personal development

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## PROJECTS

### LaptopLifeSaver — Predictive Maintenance & Fleet Management System *(AI-Powered)*

A comprehensive platform for monitoring laptop fleet health and reliability for educational institutions in Nyanza District. Combines a Python-based system monitoring agent with a real-time React dashboard and ML prediction engine.

- Built a lightweight Python agent that collects real-time hardware telemetry: CPU temperature, RAM usage, disk health, and battery status — with offline buffering for network resilience
- Implemented a machine learning prediction engine to generate health scores (0–100) and classify device status (green/yellow/red) based on historical telemetry
- Developed a real-time React dashboard with fleet overview, S.M.A.R.T. disk failure monitoring, browser push alerts, and remote maintenance action triggers
- Integrated Supabase real-time hooks for live data streaming from agent to dashboard with PostgreSQL row-level security (RLS)
- Packaged the monitoring agent as a Windows executable (.exe) with auto-installation to C:\Program Files\

**Stack:** React 19, Vite, Tailwind CSS, Python 3.10, psutil, wmi, Supabase (PostgreSQL, Realtime, Auth), PLpgSQL, GitHub Actions

### Abelus — Unified Commerce & Business Management Platform *(Enterprise-grade)*

A multi-tenant, full-stack commerce ecosystem bridging digital marketplaces and physical retail. Supports consumers, sellers, and business administrators with distinct role-based experiences.

- Built a hybrid Point of Sale (POS) system with mandatory shift enforcement, cash reconciliation, and automated expense tracking
- Implemented a real-time multi-vendor marketplace with product search, multi-seller cart, and payment gateway integration
- Developed a professional PDF financial reporting engine generating audit-ready documents with dynamic dashboards
- Engineered an Abonne (subscription/credit) module with full transaction history and staff accountability tracking
- Applied RBAC, JWT authentication, and Passport.js for secure multi-tenant access control

**Stack:** React 19, Tailwind CSS, Node.js (Express 5), PostgreSQL, Supabase, Prisma ORM, Cloudinary, PDFKit, Chart.js, Handlebars

## TECHNICAL SKILLS

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**Frontend:** React 19, Vite, Tailwind CSS, HTML5, CSS3, JavaScript (ES6+), Chart.js, React Router

**Backend:** Node.js, Express 5, Python 3, REST API design, JWT, Passport.js, RBAC

**Databases:** PostgreSQL, Supabase, Prisma ORM, MySQL, MongoDB, PLpgSQL

**ML & Systems:** Python (psutil, wmi, ML libraries), predictive modelling, real-time telemetry, system monitoring

**Tools & Platforms:** Git, GitHub, GitHub Actions, VS Code, Postman, Vercel, Cloudinary, PDFKit

**Soft Skills:** Self-directed learning, Problem-solving, Attention to detail, Teamwork, Communication

## LANGUAGES

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- Kinyarwanda — Native
- English — Professional working proficiency

## REFERENCES

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*Available upon request.*